

Brief History of the Internet

The Internet has its roots in the 1960s when the U.S. Department of Defense developed ARPANET, a network that allowed multiple computers to communicate on a single network. Over the years, the technology evolved, leading to the development of protocols that allowed different networks to interconnect.

- **1970s:** The Transmission Control Protocol (TCP) and Internet Protocol (IP) were developed, forming the basis of Internet communication.
- **1980s:** The Domain Name System (DNS) was introduced, allowing users to access websites using human-readable names instead of numerical IP addresses.
- **1990s:** The World Wide Web was created by Tim Berners-Lee, which popularized the Internet and made it accessible to the general public. The first web browser, Mosaic, was released, leading to a rapid expansion of web content.
- **2000s and beyond:** The Internet continued to evolve with the introduction of broadband, mobile Internet, and social media, leading to the interconnected world we know today.

How Does the Internet Work?

The Internet is a global network of interconnected computers that communicate using standardized protocols. Here's a simplified overview of how it works:

1. **Data Transmission:** Data is broken down into packets, which are sent over various routes through the network.
2. **Protocols:** Protocols like TCP/IP govern how data is transmitted and received. TCP ensures reliable transmission, while IP handles addressing and routing.
3. **Routers and Switches:** These devices direct packets to their destination by determining the best path through the network.
4. **End Devices:** Computers, smartphones, and servers are the endpoints that send and receive data.

Internet Protocol; HTTP

- **Internet Protocol (IP):** This is the primary protocol for sending data across the Internet. It provides addressing and routing information so that packets can reach their destination.
- **HTTP (Hypertext Transfer Protocol):** This is an application layer protocol used for transmitting hypertext (web pages) over the Internet. It defines how messages are formatted and transmitted, and how web servers and browsers should respond to various commands.

Domain Names; Domain Name Service (DNS) Servers

- **Domain Names:** These are human-readable addresses (like `www.example.com`) that correspond to IP addresses. They make it easier for users to access websites without remembering numerical addresses.
- **Domain Name Service (DNS):** This is a system that translates domain names into IP addresses. When you enter a domain name in your browser, a DNS server resolves it to the corresponding IP address, allowing your browser to locate the web server hosting the site.

HTTP Protocols

Difference between HTTP 1.0, HTTP 1.1, and HTTP 2.0

- **HTTP 1.0:** The original version, which established the basic structure of HTTP. It is stateless and does not support persistent connections.
- **HTTP 1.1:** Introduced in 1999, it added features like persistent connections (keep-alive), chunked transfer encoding, and additional caching mechanisms. It allows multiple requests and responses to be sent over a single connection.
- **HTTP 2.0:** Released in 2015, it focuses on performance improvements. It uses multiplexing (multiple streams over a single connection), header compression, and prioritization of requests, leading to faster page loads.

Methods – GET, POST, HEAD, PUT, DELETE, etc.

- **GET:** Requests data from a specified resource. It is safe and idempotent, meaning it does not change the state of the server.
- **POST:** Sends data to the server to create or update a resource. It is not idempotent, meaning multiple identical requests can have different effects.
- **HEAD:** Similar to GET, but it retrieves only the headers of a resource, not the body. Useful for checking if a resource exists.
- **PUT:** Sends data to the server to update a resource. It is idempotent, meaning multiple identical requests will have the same effect.
- **DELETE:** Requests the server to delete a specified resource.

Status Codes

HTTP status codes are issued by a server in response to a client's request. They indicate the outcome of the request:

- **200 OK:** The request was successful.
- **404 Not Found:** The requested resource could not be found.
- **500 Internal Server Error:** The server encountered an unexpected condition.

Stateless Nature of the Protocol and HTTP Session

HTTP is a stateless protocol, meaning each request from a client to a server is treated as an independent transaction. The server does not retain any information about previous requests. However, sessions can be managed using cookies or tokens to maintain state across multiple requests.

HTTPS

HTTPS (Hypertext Transfer Protocol Secure) is the secure version of HTTP. It uses encryption protocols like SSL/TLS to secure the data transmitted between the client and the server. This ensures that sensitive information, such as login credentials and payment details, is protected from eavesdropping and tampering during transmission.



Architecture of the Web

The architecture of the web consists of various components that work together to deliver content to users. Key elements include:

- **Clients:** Devices like computers and smartphones that request and display web content.
- **Web Servers:** These store, process, and deliver web pages to clients. They respond to requests made by clients using HTTP or HTTPS.
- **Databases:** Many web applications rely on databases to store and retrieve data dynamically. They work in conjunction with web servers to provide content.
- **Content Delivery Networks (CDNs):** These are distributed networks of servers that deliver web content to users based on their geographic location, improving load times and reducing latency.

Web Servers – IIS, Apache Server

- **IIS (Internet Information Services):** A web server created by Microsoft for use with the Windows operating system. It supports various protocols, including HTTP, HTTPS, FTP, and more. IIS is known for its integration with other Microsoft products and services.
- **Apache Server:** An open-source web server software that is widely used across different operating systems. It is known for its flexibility, security features, and extensive module support, allowing users to customize its functionality to meet specific needs.